IFFT BLOCK

Verification Plan

DDS for 5G/6G Cellular Network Calibration and Ranging

Faculty of Engineering, Ain Shams University | Sponsor: Analog Devices, Inc. (ADI)

Field	Value
Document Title	IFFT Block Verification Plan
Project	DDS for 5G/6G Cellular Network Calibration and Ranging
Author	Mohamed Hussein
Supervisor	Eng. Mohamed El-Adawy
Institution	Faculty of Engineering, Ain Shams University
Sponsor	Analog Devices, Inc. (ADI)
Tool	pyuvm + cocotb + Icarus Verilog
Language	Python (pyuvm), Verilog RTL
Document Version	4.0 — FINAL
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1. Introduction

This document describes the verification plan for the 4096-point Inverse Fast Fourier Transform (IFFT) block, which forms a critical component of the Transmit (TX) path in the DDS-based 5G/6G waveform generation system. The IFFT converts frequency-domain subcarrier data into a time-domain OFDM symbol, making its numerical correctness essential for system-level waveform fidelity.

The verification methodology is based on the Universal Verification Methodology (UVM) framework implemented in Python using the pyuvm library, co-simulated with the Verilog RTL via cocotb and Icarus Verilog. A 100% bit-true Python golden model (radix2_dif_ifft_fixed) that exactly mirrors the fixed-point arithmetic of the RTL is used as the reference for scoreboard comparison.

Version 4.0 is the final version of this document, reflecting the completion of Round 6 — the last verification round. All individual tests and small regression group tests have been run with both multi_frame modes. 100% functional coverage was achieved in the regression test. One known issue (BUG-003) was identified in the random test where the first output sample is incorrect while all subsequent samples pass. The random test is the only failing test; all other tests and regressions pass.

2. DUT Overview

2.1 Module Hierarchy

Module File	Description
fft_4096_top.v	Top-level IFFT wrapper — primary DUT interface
fft_stage.v	Single butterfly stage with twiddle ROM instantiation (12 instances)
butterfly.v	Radix-2 DIF butterfly — complex add/sub with 16-bit wrap
multiplier.v	Complex multiplier — Q2.14 twiddle, nearest-integer rounding via bit[13]
controlunit.v	valid_in / valid_out pipeline control — N-1 cycle delay shift register
delayfeedback.v	Pipeline delay chain — holds intermediate stage data
twiddlerom.v	Pre-computed twiddle factor ROM (Q2.14 signed 16-bit, generated offline)

2.2 Interface Signals

Signal	Dir	Width	Description

clk	In	1	System clock — 491.52 MHz (period ≈ 2034.5 ps)
rst_n	In	1	Active-low synchronous reset — clears all pipeline registers
valid_in	In	1	Asserted high for each valid frequency-domain input sample
in_real	In	16	Real part of frequency-domain input (signed Q11.5)
in_imag	In	16	Imaginary part of frequency-domain input (signed Q11.5)
multi_frame	In	1	Frame-overlap control — see Section 2.4
valid_out	Out	1	Asserted high for each valid time-domain output sample
out_real	Out	16	Real part of time-domain IFFT output (signed Q11.5)
out_imag	Out	16	Imaginary part of time-domain IFFT output (signed Q11.5)

2.3 Key Design Parameters

Parameter	Value	Notes
FFT Size (N)	4096	12 pipeline stages (log2 4096)
Word Length (WL)	16 bits	Signed fixed-point data path
Data Format	Q11.5	5 fractional bits — scale factor = 32
Twiddle Format	Q2.14	Scale = 16384, stored in twiddlerom.v ROM
Pipeline Latency	N−1 = 4095 cycles	From first valid_in to first valid_out
valid_out timing	Cycle 4095 (0-idx)	One cycle before N — confirmed intended (Round 3)
Clock Frequency	491.52 MHz	Period ≈ 2034.5 ps
IFFT mechanism	Twiddle conjugation	RTL negates twiddle_rom_img — no circular shift needed

Comparison tolerance	±2 LSBs	Accounts for rounding divergence across 12 stages
DC overflow threshold	in > 0.46875 (0x0F in Q11.5)	Confirmed BUG-002 — see Section 6

2.4 multi_frame Signal

The multi_frame signal controls whether the IFFT pipeline operates in overlapped or non-overlapped mode between consecutive frames.

multi_frame value	Behaviour
True (1)	Drive input frames one after another without waiting (overlapped mode). Second frame input begins before first frame output completes. Used in back-to-back streaming scenarios.
False (0)	Wait until each frame finishes its complete output before driving the next frame (non-overlapped mode). Used when frame isolation is required.

All individual sequence tests are run both with multi_frame=True and multi_frame=False. The functional regression test (ifft_functional_test) has been confirmed passing in both modes.

3. Verification Environment

3.1 Environment Hierarchy

The verification environment is part of a larger configurable TX-path testbench. For IFFT block-level testing, the IFFT agent is set to UVM_ACTIVE and all other agents are UVM_PASSIVE. VERIF_MODE is set to IFFT in the ConfigDB.

Component	File	Role
ifft_base_test	ifft_base_test.py	Activates ifft_agent, sets VERIF_MODE=IFFT in ConfigDB
ifft_item	ifft_item.py	UVM sequence item — carries all DUT signals, CRV randomization
Individual seqs	ifft_*_seq.py (16)	One sequence file per test — see Section 4
Regression seqs	ifft_sequences.py	Grouping sequences for functional/stress/frequency/regression tests
Scoreboard	ifft_scoreboard.py	Compares DUT output vs bit-true golden model, ±2 LSB tolerance
Subscriber	ifft_subscriber.py	Functional coverage — 8 cover points + 1 CoverCross
Golden Model	ifft_golden_model.py	radix2_dif_ifft_fixed — 100% bit-true Python translation of RTL

3.2 Sequence Item — ifft_item

Field	Type	Randomized	Description
rst_n	bit {0,1}	Yes	Active-low synchronous reset
valid_in	bit {0,1}	Yes	Input valid handshake
in_real	int [-32768,32767]	Yes	Signed Q11.5 real frequency-domain input
in_imag	int [-32768,32767]	Yes	Signed Q11.5 imaginary frequency-domain input
multi_frame	bit {0,1}	Yes	Frame-overlap control (True=overlapped, False=non-overlapped)
valid_out	bit	No	Captured by monitor from DUT — not driven
out_real	int	No	Captured by monitor from DUT — not driven
out_imag	int	No	Captured by monitor

			from DUT — not driven

3.3 Golden Model

The scoreboard uses radix2_dif_ifft_fixed() from ifft_golden_model.py as its reference. Fixed-point arithmetic is identical to the RTL:

- Twiddle factors in Q2.14: round(exp(-j2πk/m) × 16384), clamped to [-32768, 32767]
- IFFT conjugation: W_im negated — same as RTL negating twiddle_rom_img
- Butterfly outputs wrapped to 16-bit signed via wrap_nbit() — mirrors Verilog wire
- Multiplier truncation: floor(product/16384) + (product//8192)%2 — nearest-integer rounding via bit[13]
- Output reordered by bit-reversal permutation identical to MATLAB bitrevorder()
- Comparison tolerance: ±2 LSBs — absorbs rounding divergence across 12 stages

3.4 Functional Coverage — Subscriber

CP ID	Signal	Bins	Goal
CP1	valid_in	0 (de-asserted), 1 (asserted)	Both handshake states seen
CP2	valid_out	0 (de-asserted), 1 (asserted)	Output valid both states observed
CP7	rst_n	0 (reset active), 1 (reset inactive)	Reset exercised at least once
CP3	in_real range	min_neg / negative / zero / positive / max_pos	Full signed range of real input
CP4	in_imag range	min_neg / negative / zero / positive / max_pos	Full signed range of imag input
CP5	out_real range	min_neg / negative / zero / positive / max_pos	Full signed range of real output
CP6	out_imag range	min_neg / negative / zero / positive / max_pos	Full signed range of imag output
CP8 (Cross)	in_real × in_imag quadrant	(pos,pos) (pos,neg) (neg,pos) (neg,neg)	All four input quadrants exercised

4. Test Architecture

The test suite is structured in two tiers:

- Individual tests: each test file runs exactly one sequence file. Every sequence is exercised in isolation with both multi_frame=True and multi_frame=False.
- Group / regression tests: test files that run multiple sequences to cover a theme (functional correctness, frequency sweep, stress, etc.) or the full regression.

4.1 Individual Test Files — One Sequence Each

Test File	Sequence File	Stimulus Pattern	multi_frame tested	Status
ifft_reset_test.py	ifft_reset_seq.py	rst_n = 0	Both	PASS
ifft_allzero_test.py	ifft_allzero_seq.py	All bins 0+0j	Both	PASS
ifft_impulse_test.py	ifft_impulse_seq.py	X[0]=1+0j, rest=0	Both	PASS
ifft_dc_test.py	ifft_dc_seq.py	All bins = 1+0j	Both	PASS
ifft_singletone_test.py	ifft_singletone_seq.py	Unit phasor at one bin k	Both	PASS
ifft_twotone_test.py	ifft_twotone_seq.py	Bins k=10 and k=200 active	Both	PASS
ifft_maxpositive_test.py	ifft_maxpositive_seq.py	in_real = +32767 every bin	Both	PASS
ifft_maxnegative_test.py	ifft_maxnegative_seq.py	in_real = -32768 every bin	Both	PASS
ifft_alternating_test.py	ifft_alternating_seq.py	+32767 / -32768 alternating	Both	PASS
ifft_halfnyquist_test.py	ifft_halfnyquist_seq.py	Only bin N/2 = 2048	Both	PASS
ifft_ramp_test.py	ifft_ramp_seq.py	in_real ramps 0 → 0.9	Both	PASS
ifft_random_test.py	ifft_random_seq.py	Free random in_real/in_imag	Both	FAIL
ifft_burst_test.py	ifft_burst_seq.py	valid_in=1 for 512 samples	Both	PASS
ifft_backtoback_test.py	ifft_backtoback_seq.py	Two frames, no gap	Both	PASS
ifft_fullregression_test.py	ifft_fullregression_seq.py	All 15 sequences in order	Both	PASS

4.2 Group / Regression Test Files

Test File	Sequences Run	Role	Status
ifft_functional_test.py	allzero → impulse → dc → singletone → twotone	Mini regression — core IFFT math correctness. Tested with multi_frame=True and multi_frame=False.	**PASS**
ifft_frequency_test.py	singletone k=1/100/2047 → twotone → halfnyquist	Frequency-axis sweep — twiddle ROM at low, mid, high, Nyquist bins	**PASS**
ifft_stress_test.py	maxpositive → maxnegative → alternating → ramp	Datapath limits — saturation, carry/borrow, full dynamic range	**PASS**
ifft_random_test.py	random (small regression — single random frame)	Broad coverage — reproducible random stimulus. First output sample mismatch — BUG-003.	**FAIL**
ifft_regression_test.py	All 15 individual sequences in order	Full nightly regression — 100% functional coverage achieved. Passes with multi_frame=True and False.	**PASS**

5. Sequence Library

Each sequence sends 2N = 8192 clock cycles: the first N cycles drive the input frame with valid_in=1, the second N cycles idle with valid_in=0 (multi_frame=False) or immediately start the next frame (multi_frame=True). All sequences constrain only rst_n and valid_in via randomize_with(), then directly assign in_real and in_imag to avoid CRV solver timeout on the 65536-element data domain.

#	Sequence	Input Pattern	Expected Output	Purpose
1	ifft_reset_seq	rst_n = 0	valid_out = 0, all registers clear	Baseline reset sanity
2	ifft_allzero_seq	All bins 0+0j	All-zero output	Zero-input baseline
3	ifft_impulse_seq	X[0]=1+0j, rest=0	Constant 1/N for all n	DC-bin path + normalisation
4	ifft_dc_seq	All bins = 1+0j	Spike at n=0, zeros elsewhere	Max constructive butterfly addition
5	ifft_singletone_seq	Unit phasor at bin k	Pure sinusoid at f=k/N	Twiddle ROM accuracy per bin
6	ifft_twotone_seq	Bins k=10 and k=200	Sum of two sinusoids	Butterfly linearity check
7	ifft_maxpositive_seq	in_real = +32767 every bin	Golden model output	Positive saturation stress
8	ifft_maxnegative_seq	in_real = -32768 every bin	Golden model output	Negative saturation stress
9	ifft_alternating_seq	+32767 / -32768 alternating	Golden model output	Carry/borrow propagation stress
10	ifft_halfnyquist_seq	Only bin N/2 = 2048	Alternating ±1/N	Nyquist twiddle W=−1 corner case
11	ifft_ramp_seq	in_real ramps 0 → 0.9	Golden model output	Full dynamic range
12	ifft_random_seq	Free random in_real/in_imag	Golden model output	Broad functional coverage
13	ifft_burst_seq	valid_in=1 for 512 samples	No spurious valid_out	Partial-frame handling
14	ifft_backtoback_seq	Two frames, no gap	Two correct output frames	Pipeline frame overlap
15	ifft_fullregression_seq	All 14 sequences in order	All pass golden model	Full regression in one sequence

6. Verification Progress

6.1 Round-by-Round Summary

Round	Activity	Outcome	Status
Round 1	Environment setup — hierarchy, ifft_item, sequences, scoreboard, subscriber, golden model, Makefile	All components compile. Environment instantiates cleanly.	**PASS**
Round 2	ifft_functional_test — allzero_sequence ifft_functional_test — impulse_sequence	allzero: 0 errors. impulse: Scoreboard detects valid_out at cycle 4095 instead of 4096 — logged as BUG-001.	**PASS**
Round 3	impulse_sequence (modified — last input changed to non-impulse value)	Output changes when last sample changes. RTL correctly processes all 4096 inputs. BUG-001 closed as By Design. N-1 latency is intended.	**PASS**
Round 4	Architecture refactored: every sequence moved to its own file and test. multi_frame signal added to DUT and seq-item. ifft_reset, allzero, impulse, dc, singletone, twotone, maxpositive, maxnegative tests run (multi_frame=True and False). ifft_functional_test confirmed passing in both multi_frame modes.	8 individual tests pass. ifft_functional_test passes (both multi_frame modes). BUG-002 discovered in ifft_dc_test — see Section 6.3.	**PASS**
Round 5	ifft_alternating_test, ifft_halfnyquist_test, ifft_ramp_test run (multi_frame=True and False). ifft_frequency_test (group) run in both multi_frame modes. ifft_stress_test (group) run in both multi_frame	All 5 individual tests pass. ifft_frequency_test passes in both multi_frame modes. ifft_stress_test passes in both multi_frame modes. No new bugs found.	**PASS**

	modes.		
Round 6 (Final)	All remaining individual tests run: ifft_random, ifft_burst, ifft_backtoback, ifft_fullregression (multi_frame=True and False). ifft_random_test (group) run. ifft_regression_test (full nightly) run in both multi_frame modes. 100% functional coverage achieved in regression test.	ifft_burst, ifft_backtoback, ifft_fullregression: all PASS. ifft_regression_test: PASS in both multi_frame modes — 100% coverage. ifft_random_test: FAIL — first output sample is wrong, all subsequent samples correct. Logged as BUG-003. All other tests and group regressions: PASS.	**PASS**

6.2 Bug Log

Bug ID	Round	Description	Root Cause	Resolution	Status
BUG-001	Round 2	valid_out asserts at cycle 4095 (0-indexed) — one cycle earlier than expected N=4096	Scoreboard expected valid_out after N cycles. RTL control unit's N-1 shift register asserts after N-1 cycles.	Round 3: last input modified — output changed, proving RTL uses all 4096 inputs. N-1 latency is intended. Scoreboard updated.	CLOSED By Design
BUG-002	Round 4	DC input of 1+0j (all bins) produces overflow — output at n=0 is 0 instead of expected N=4096. Overflow occurs for any input > 0.46875 (0x0F in Q11.5).	The IFFT does not divide by N. All N bins each contributing 1 produces a constructive sum of 4096 at n=0, which overflows the 16-bit Q11.5 range.	Under investigation. DC input must be scaled below 0.46875 per bin. Designer to be consulted on whether N-normalisation should be added.	OPEN
BUG-003	Round 6	ifft_random_test: the very first output sample (index 0) of the random frame is incorrect. All N-1 subsequent	Under investigation. The first output sample emerges at the pipeline boundary	Under investigation. All other sequences and the full regression test pass —	OPEN

		output samples match the golden model within tolerance.	(cycle N-1). Suspected cause: the pipeline registers may not be fully flushed from the previous sequence before the first valid_out assertion, causing the first output to carry residual state.	the issue is isolated to the first output sample of a random-data frame following another sequence. Designer to be consulted.	

6.3 BUG-001 Detail — valid_out Timing Investigation (Round 3)

During Round 2, the scoreboard flagged a cycle-alignment mismatch: valid_out was observed high at cycle 4095 (0-indexed from the first valid_in), while the golden model expected it at cycle 4096.

Experiment: The impulse_sequence was modified so that the last sample (index 4095) was changed to a different value. The output changed, confirming the RTL correctly processes all 4096 inputs.

Conclusion: BUG-001 is closed as By Design. The RTL is correct. The N-1 cycle latency is the intended behaviour of the control unit shift register.

6.4 BUG-002 Detail — DC Overflow (Round 4)

During ifft_dc_test, the input was set to 1+0j for all N=4096 frequency bins. The expected time-domain output at n=0 is the sum of all N bins = 4096 (since the RTL does not perform the 1/N normalisation). This value far exceeds the 16-bit Q11.5 signed range of [-32768, +32767] integers, causing overflow and producing 0 at the output.

The overflow threshold was determined experimentally: any DC bin amplitude greater than 0.46875 (which is 0x0F in raw Q11.5 integer representation, corresponding to approximately 0.5 considering Q11.5 precision) causes the output at n=0 to overflow and wrap to 0.

- Safe DC amplitude range: in_real per bin ≤ 0x0F (0.46875 float) — output stays within 16-bit range
- Unsafe DC amplitude: in_real per bin = 0x20 (1.0 float, i.e. full scale) — output overflows to 0
- Root cause: IFFT output = sum of all bins × twiddle products. For all-ones input, constructive addition at n=0 produces N × amplitude, which overflows Q11.5 for amplitude > 1/N ≈ 0.000244 × N = 1.0 per bin

Status: OPEN — under discussion with designer. Scoreboard tolerance updated to flag this as a known overflow condition rather than a mismatch error.

6.5 BUG-003 Detail — Random Test First Sample Mismatch (Round 6)

During ifft_random_test, the scoreboard detected that the very first output sample (index 0, corresponding to the first valid_out assertion) does not match the golden model. All N-1 = 4095 subsequent output samples match correctly within tolerance.

This issue does not appear in any other sequence — the regression test passes completely, including all 15 sequences. The random test is the only failing test.

- Samples 1 through 4095: all match golden model — PASS
- Sample 0 (first valid_out): mismatch between DUT and golden model — FAIL
- Suspected cause: residual pipeline state from the preceding sequence bleeds into the first output sample of the random frame, because the random input data does not have a predictable leading pattern that would mask the residual
- Note: the regression test (which includes the random sequence among 14 others) still passes — suggesting the issue may be specific to the sequence ordering or the frame boundary condition in the standalone random test

Status: OPEN — isolated to first output sample of random frame. All other tests and the full regression pass. Designer consultation pending.

7. Coverage Goals

Coverage Type	Metric	Target	Current
Functional (CP1–CP7)	Bin hit rate	100%	100% — achieved in regression test
Functional (CP8 Cross)	Quadrant hit rate	100%	100% — achieved in regression test
Scoreboard — out_real	Error count	0 errors	0 errors (all tests except random first sample — BUG-003)
Scoreboard — out_imag	Error count	0 errors	0 errors (all tests except random first sample — BUG-003)
Individual test coverage	Tests run / total	15 / 15	15 / 15 — complete (14 PASS, 1 FAIL — random)
Group test coverage	Tests run / total	5 / 5	5 / 5 — complete (4 PASS, 1 FAIL — random)
multi_frame coverage	Both modes tested	True + False on all tests	True + False on all 15 tests and all 5 group tests

8. Outstanding Issues

Verification is complete. All tests have been run. Two open bugs remain for designer consultation:

1. BUG-002 (DC overflow) — DC input > 0.46875 per bin causes output overflow at n=0. Designer to decide whether 1/N normalisation should be added to the RTL output stage or whether input pre-scaling is the responsibility of the upstream block.
2. BUG-003 (Random first sample mismatch) — The first valid_out sample of the random sequence is incorrect. All 4095 subsequent samples pass. Suspected residual pipeline state at the frame boundary. Designer consultation pending.

9. Makefile Targets

make target	Test file	Sequences executed
make reset	ifft_reset_test.py	ifft_reset_seq
make allzero	ifft_allzero_test.py	ifft_allzero_seq

make impulse	ifft_impulse_test.py	ifft_impulse_seq
make dc	ifft_dc_test.py	ifft_dc_seq
make singletone	ifft_singletone_test.py	ifft_singletone_seq
make twotone	ifft_twotone_test.py	ifft_twotone_seq
make maxpositive	ifft_maxpositive_test.py	ifft_maxpositive_seq
make maxnegative	ifft_maxnegative_test.py	ifft_maxnegative_seq
make alternating	ifft_alternating_test.py	ifft_alternating_seq
make halfnyquist	ifft_halfnyquist_test.py	ifft_halfnyquist_seq
make ramp	ifft_ramp_test.py	ifft_ramp_seq
make random	ifft_random_test.py	ifft_random_seq (small regression — single random frame)
make burst	ifft_burst_test.py	ifft_burst_seq
make backtoback	ifft_backtoback_test.py	ifft_backtoback_seq
make fullregression	ifft_fullregression_test.py	ifft_fullregression_seq (all 14)
make functional	ifft_functional_test.py	allzero → impulse → dc → singletone → twotone
make frequency	ifft_frequency_test.py	singletone k=1/100/2047 → twotone → halfnyquist
make stress	ifft_stress_test.py	maxpositive → maxnegative → alternating → ramp
make regression	ifft_regression_test.py	All 16 sequences in one run
make all	All individual tests in order	Runs all 15 with clean_sim between each

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End of Document — IFFT Verification Plan v4.0 (FINAL) | Ain Shams University | ADI | April 2026

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Document Title	IFFT Block Verification Plan
Project	DDS for 5G/6G Cellular Network Calibration and Ranging
Author	Mohamed Hussein
Institution	Faculty of Engineering, Ain Shams University
Sponsor	Analog Devices, Inc. (ADI)
Tool	pyuvm + cocotb + Icarus Verilog
Language	Python (pyuvm), Verilog RTL
Document Version	1.0
Date	April 2026
Status	In Progress — Round 3 completed

1. Introduction

This document describes the verification plan for the 4096-point Inverse Fast Fourier Transform (IFFT) block, which forms a critical component of the Transmit (TX) path in the DDS-based 5G/6G waveform generation system. The IFFT converts frequency-domain subcarrier data into a time-domain OFDM symbol, making its numerical correctness essential for system-level waveform fidelity.

The verification methodology is based on the Universal Verification Methodology (UVM) framework implemented in Python using the pyuvm library, co-simulated with the Verilog RTL via cocotb and Icarus Verilog. A 100% bit-true Python golden model (radix2_dif_ifft_fixed) that exactly mirrors the fixed-point arithmetic of the RTL — including Q2.14 twiddle factor quantisation, 16-bit wrapping, and nearest-integer rounding — is used as the reference for scoreboard comparison.

2. DUT Overview

2.1 Module Hierarchy

Module File	Description
fft_4096_top.v	Top-level IFFT wrapper — primary DUT interface
fft_stage.v	Single butterfly stage with twiddle ROM instantiation (12 instances)
butterfly.v	Radix-2 DIF butterfly — complex add/sub with 16-bit wrap
multiplier.v	Complex multiplier — Q2.14 twiddle, nearest-integer rounding via bit[13]
controlunit.v	valid_in / valid_out pipeline control — N-1 cycle delay shift register
delayfeedback.v	Pipeline delay chain — holds intermediate stage data between butterfly groups
twiddlerom.v	Pre-computed twiddle factor ROM (Q2.14 signed 16-bit, generated offline)

2.2 Interface Signals

Signal	Dir	Width	Description
clk	In	1	System clock — 491.52 MHz (period ≈ 2034.5 ps)
rst_n	In	1	Active-low synchronous reset — clears all pipeline registers

valid_in	In	1	Asserted high for each valid frequency-domain input sample
in_real	In	16	Real part of frequency-domain input (signed Q11.5)
in_imag	In	16	Imaginary part of frequency-domain input (signed Q11.5)
valid_out	Out	1	Asserted high for each valid time-domain output sample
out_real	Out	16	Real part of time-domain IFFT output (signed Q11.5)
out_imag	Out	16	Imaginary part of time-domain IFFT output (signed Q11.5)

2.3 Key Design Parameters

Parameter	Value	Notes
FFT Size (N)	4096	12 pipeline stages (log2 4096)
Word Length (WL)	16 bits	Signed fixed-point data path
Data Format	Q11.5	5 fractional bits — scale factor = 32
Twiddle Format	Q2.14	Scale = 16384, stored in twiddlerom.v ROM
Pipeline Latency	N−1 = 4095 cycles	From first valid_in to first valid_out
valid_out timing	Cycle 4095 (0-idx)	One cycle before N — confirmed intended (Round 3)
Clock Frequency	491.52 MHz	Period ≈ 2034.5 ps
IFFT mechanism	Twiddle conjugation	RTL negates twiddle_rom_img — no circular shift needed
Comparison tolerance	±2 LSBs	Accounts for rounding divergence across 12 stages

3. Verification Environment

3.1 Environment Hierarchy

The verification environment is part of a larger configurable TX-path testbench that can target the DDS, FFT, IFFT, or full TOP blocks. For IFFT block-level testing, the IFFT agent is set to UVM_ACTIVE and all other agents (top_agent, dds_agent, fft_agent) are set to UVM_PASSIVE via the ConfigDB. The VERIF_MODE key is set to "IFFT" so monitors and scoreboards know which DUT is under test.

Component	File	Role
ifft_base_test	ifft_base_test.py	Activates ifft_agent, sets VERIF_MODE=IFFT in ConfigDB
ifft_item	ifft_item.py	UVM sequence item — carries all DUT signals, CRV randomization
ifft_sequences	ifft_sequences.py	17 sequence classes (14 directed + reset + burst + regression)
Scoreboard	ifft_scoreboard.py	Receives items, runs golden model, compares DUT vs reference
Subscriber	ifft_subscriber.py	Functional coverage — 8 cover points + 1 CoverCross
Golden Model	ifft_golden_model.py	radix2_dif_ifft_fixed — 100% bit-true Python translation of RTL

3.2 Sequence Item — ifft_item

Field	Type	Randomized	Description
rst_n	bit {0,1}	Yes	Active-low synchronous reset
valid_in	bit {0,1}	Yes	Input valid handshake
in_real	int [-32768,32767]	Yes	Signed Q11.5 real frequency-domain input
in_imag	int [-32768,32767]	Yes	Signed Q11.5 imaginary frequency-domain input
valid_out	bit	No	Captured by monitor from DUT — not driven
out_real	int	No	Captured by monitor from DUT — not driven
out_imag	int	No	Captured by monitor

			from DUT — not driven

3.3 Golden Model

The scoreboard uses radix2_dif_ifft_fixed() from ifft_golden_model.py as its reference. This function is a line-by-line Python translation of the MATLAB golden model. The following fixed-point arithmetic is identical to the RTL:

- Twiddle factors generated as Q2.14 integers: round(exp(-j2πk/m) × 16384), clamped to [-32768, 32767]
- IFFT twiddle conjugation: W_im negated (same as RTL negating twiddle_rom_img)
- Butterfly add/sub outputs wrapped to 16-bit signed using wrap_nbit() — mirrors Verilog wire wrap
- Multiplier truncation: floor(product / 16384) + (product // 8192) % 2 — nearest-integer rounding via bit[13]
- Output reordered by bit-reversal permutation identical to MATLAB bitrevorder()
- Comparison tolerance: ±2 LSBs — absorbs rounding divergence accumulated across 12 stages

3.4 Functional Coverage — Subscriber

CP ID	Signal	Bins	Goal
CP1	valid_in	0 (de-asserted), 1 (asserted)	Both handshake states seen
CP2	valid_out	0 (de-asserted), 1 (asserted)	Output valid both states observed
CP7	rst_n	0 (reset active), 1 (reset inactive)	Reset exercised at least once
CP3	in_real range	min_neg / negative / zero / positive / max_pos	Full signed range of real input
CP4	in_imag range	min_neg / negative / zero / positive / max_pos	Full signed range of imag input
CP5	out_real range	min_neg / negative / zero / positive / max_pos	Full signed range of real output
CP6	out_imag range	min_neg / negative / zero / positive / max_pos	Full signed range of imag output
CP8 (Cross)	in_real × in_imag quadrant	(pos,pos) (pos,neg) (neg,pos) (neg,neg)	All four input quadrants exercised

4. Sequence Library

Each sequence sends 2N = 8192 clock cycles total: the first N cycles drive the input frame with valid_in=1, the second N cycles idle with valid_in=0 to allow the pipeline output to emerge and be captured by the monitor. All sequences use randomize_with() to constrain only rst_n and valid_in, then directly assign in_real and in_imag — this avoids the CRV solver hanging on the 65536-element domain of the data signals.

#	Sequence Class	Input Pattern	Expected Output	Purpose
1	reset_sequence	rst_n = 0	valid_out = 0, registers clear	Baseline reset sanity
2	allzero_sequence	All bins 0+0j	All-zero output	Zero-input baseline
3	impulse_sequence	X[0]=1+0j, rest=0	Constant 1/N for all n	DC-bin path + normalisation
4	dc_sequence	All bins = 1+0j	Spike at n=0, zeros elsewhere	Max constructive butterfly addition
5	singletone k=1	Unit phasor at bin 1	Pure sinusoid at f=1/N	Low-freq twiddle ROM entries
6	singletone k=100	Unit phasor at bin 100	Pure sinusoid at f=100/N	Mid-freq twiddle ROM entries
7	singletone k=2047	Unit phasor at bin 2047	Pure sinusoid at f=2047/N	High-freq, just below Nyquist
8	twotone_sequence	Bins k=10 and k=200	Sum of two sinusoids	Butterfly linearity
9	maxpositive_sequence	in_real = +32767 every bin	Golden model output	Positive saturation stress
10	maxnegative_sequence	in_real = -32768 every bin	Golden model output	Negative saturation stress
11	alternating_sequence	+32767 / -32768 alternating	Golden model output	Carry/borrow propagation
12	halfnyquist_sequence	Only bin N/2 = 2048	Alternating ±1/N	Nyquist twiddle W=-1 corner
13	ramp_sequence	in_real ramps 0 → 0.9	Golden model output	Full dynamic range
14	random_sequence	Free random in_real/in_imag	Golden model output	Broad functional coverage
15	burst_sequence	valid_in=1 for 512 samples	No spurious valid_out	Partial-frame handling
16	resetmidframe_sequence	Reset at sample N/2	Clean output post-reset	Reset recovery
17	backtoback_sequence	Two frames, no	Two correct	Pipeline overlap

		gap	output frames	

5. Test Plan

Test File	Sequences Run	Corner Cases Targeted
ifft_reset_test.py	reset_sequence	rst_n flushes all 4095 delay registers; valid_out stays 0
ifft_functional_test.py	allzero → impulse → dc → singletone k=1 → twotone	Core IFFT mathematics, zero, DC, impulse, single-tone, linearity
ifft_frequency_test.py	tone k=1/100/2047 → twotone → nyquist	Twiddle ROM at low, mid, high, and Nyquist frequency bins
ifft_stress_test.py	maxpositive → maxnegative → alternating → ramp	Overflow, saturation, carry/borrow, full Q11.5 dynamic range
ifft_random_test.py	random seed=42 → random seed=99	Broad coverage, reproducible random stimulus
ifft_pipeline_test.py	backtoback → burst	Pipeline frame overlap; partial-frame valid_out handling
ifft_reset_recovery_test.py	reset → resetmidframe → impulse	All delay registers flush; no residual state survives reset
ifft_regression_test.py	fullregression_sequence (all 17 sequences)	Full regression — maximum coverage in one simulation

6. Verification Progress

6.1 Progress Summary

Round	Test / Sequence	Outcome	Status
Round 1	Environment setup — hierarchy, seq-item, sequences, scoreboard, subscriber, Makefile	All components compile. Environment instantiates cleanly.	**PASS**
Round 2	ifft_functional_test — allzero_sequence	All-zero input produces all-zero output. Scoreboard reports 0 errors.	**PASS**
Round 2	ifft_functional_test — impulse_sequence	Scoreboard detects valid_out asserts at cycle 4095 (0-indexed). Mismatch logged as BUG-001.	**PASS**
Round 3	impulse_sequence (modified — last input changed)	Output changes when last sample changes. RTL correctly uses all 4096 inputs. BUG-001 closed as By Design.	**PASS**
—	ifft_functional_test — dc_sequence	Not yet run	**NOT RUN**
—	ifft_functional_test — singletone k=1	Not yet run	**NOT RUN**
—	ifft_functional_test — twotone_sequence	Not yet run	**NOT RUN**
—	ifft_frequency_test	Not yet run	**NOT RUN**
—	ifft_stress_test	Not yet run	**NOT RUN**
—	ifft_random_test	Not yet run	**NOT RUN**
—	ifft_pipeline_test	Not yet run	**NOT RUN**
—	ifft_reset_recovery_test	Not yet run	**NOT RUN**
—	ifft_regression_test	Not yet run	**NOT RUN**

6.2 Bug Log

Bug ID	Round	Description	Root Cause	Resolution	Status
BUG-001	Round 2	valid_out asserts at cycle 4095 (0-indexed) —	Scoreboard expected valid_out after N	Round 3: last input sample modified — output	CLOSED By Design

		one cycle earlier than expected N=4096	cycles. RTL control unit's N-1 shift register asserts valid_out after N-1 cycles.	changed, proving RTL uses all 4096 inputs. N-1 latency is intended pipeline behaviour. Scoreboard updated.	

6.3 BUG-001 Detail — valid_out Timing Investigation

During Round 2, the scoreboard flagged a cycle-alignment mismatch: valid_out was observed high at cycle 4095 (0-indexed from the first valid_in), while the golden model expected it at cycle 4096.

Hypothesis: the RTL does not sample the 4096th input — it processes only 4095 inputs and therefore produces a valid_out one cycle early.

Experiment (Round 3): The impulse_sequence was modified so that the last sample (index 4095) was changed from the expected impulse value to a different value. The output was observed:

- If output changes → RTL uses the 4096th sample → valid_out timing is intentional
- If output does not change → RTL discards last sample → real bug

Result: the output changed when the last input was modified, confirming the RTL correctly processes all 4096 inputs. The N-1 cycle latency is the designed pipeline behaviour of the control unit shift register.

Conclusion: BUG-001 is closed as By Design. The RTL is correct.

7. Coverage Goals

Coverage Type	Metric	Target	Current
Functional (CP1–CP7)	Bin hit rate	100%	In Progress
Functional (CP8 Cross)	Quadrant hit rate	100%	In Progress
Scoreboard — out_real	Error count	0 errors	0 (allzero + impulse)
Scoreboard — out_imag	Error count	0 errors	0 (allzero + impulse)
Sequence coverage	Sequences run	17 / 17	3 / 17
Test coverage	Tests fully passing	8 / 8	1 / 8 complete

8. Remaining Work

3. Complete ifft_functional_test: run dc_sequence, singletone k=1, and twotone_sequence
4. Run ifft_frequency_test: singletone k=100, k=2047, twotone, halfnyquist
5. Run ifft_stress_test: maxpositive, maxnegative, alternating, ramp
6. Run ifft_random_test: two random frames (seed=42, seed=99)
7. Run ifft_pipeline_test: backtoback and burst sequences
8. Run ifft_reset_recovery_test: resetmidframe then clean impulse frame
9. Run ifft_regression_test: full regression all 17 sequences
10. Verify all 8 functional cover points reach 100% bin hit rate
11. Update Section 6 with new progress rounds as testing continues

9. Makefile Targets

make target	Test file	Sequences executed
make reset	ifft_reset_test.py	reset_sequence
make functional	ifft_functional_test.py	allzero → impulse → dc → singletone k=1 → twotone
make frequency	ifft_frequency_test.py	tone k=1 → k=100 → k=2047 → twotone → nyquist
make stress	ifft_stress_test.py	maxpositive → maxnegative → alternating → ramp
make random	ifft_random_test.py	random seed=42 → random seed=99
make pipeline	ifft_pipeline_test.py	backtoback → burst
make reset_recovery	ifft_reset_recovery_test.py	reset → resetmidframe → impulse

make full_regression	ifft_regression_test.py	fullregression_sequence (all 17)
make regression	All 8 tests in order	Runs all tests sequentially with clean_sim between each

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